

Evaluating Cross-Domain Transfer for Depression and Suicide Detection in Social Media Text

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Abstract

This project addresses depression detection through social media analysis using cross-domain evaluation. A supervised binary classifier is trained on the MDDL Twitter dataset and evaluated out-of-domain on a suicidal intention dataset, representing a topical domain shift (depression → suicidal ideation). Three feature representations are compared across six classifiers: TF-IDF, RoBERTa, and MentalBERT. Results show that TF-IDF fails catastrophically under domain shift (F1 drop up to 99.7%), while RoBERTa embeddings substantially improve cross-domain performance. Notably, KNN reverses from worst performer with TF-IDF (F1: 0.0009) to best with RoBERTa (F1: 0.6447), demonstrating that feature representation quality is the dominant factor in cross-domain NLP transfer.

1 Introduction

Depression affects an estimated 322 million people worldwide and is the leading cause of disability globally (Friedrich, 2017). Despite its severity, many affected individuals do not receive adequate treatment due to barriers such as social stigma and limited access to mental healthcare (Henderson et al., 2013). Social media platforms, particularly Twitter, have emerged as a valuable source of behavioural and psychological signals enabling automated detection of depression-related content (De Choudhury et al., 2013).

1.1 Motivation

Depression and suicidal ideation are clinically related but linguistically distinct mental health conditions. If a classifier trained on depression data could detect suicidal content, it would enable early intervention without requiring labelled suicidal data, which is expensive and ethically sensitive to collect (Alqahtani et al., 2025).

1.2 Problem Statement

Most existing depression detection systems are developed and evaluated within a single mental health domain, limiting their generalizability across related psychological conditions (Tavchioski et al., 2023). This project investigates cross-domain transfer from depression detection to suicidal ideation detection within Twitter data, representing a topical domain shift (depression → suicidal ideation).

1.3 Main Contributions

The main contributions of this work are:

- A systematic comparison of TF-IDF, RoBERTa, and MentalBERT across six classifiers under identical cross-domain experimental conditions.
- A vocabulary overlap analysis quantifying the linguistic gap between the two domains (40% shared top-50 words), directly explaining model failure patterns.
- Empirical evidence that general-purpose RoBERTa outperforms domain-specific MentalBERT in feature-extraction mode due to training scale advantage.
- Discovery of the KNN reversal: KNN goes from worst performer (F1: 0.0009 with TF-IDF) to best performer (F1: 0.6447 with RoBERTa).

1.4 Paper Organization

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the datasets and methodology. Section 4 presents the experimental results. Section 5 discusses and interprets the findings. Section 6 concludes the paper.

2 Related Work

Research on automated mental health detection from social media has grown significantly in recent years. [De Choudhury et al. \(2013\)](#) demonstrated that linguistic and behavioural signals extracted from Twitter can predict the onset of depression using supervised classification, establishing a foundational framework for social media-based mental health analysis.

Building on this, [Tavchioski et al. \(2023\)](#) applied and compared multiple transformer-based models — including BERT, RoBERTa, BERTweet, and MentalBERT — for depression detection across Reddit and Twitter datasets. Their work explicitly investigates cross-dataset transfer and confirms that performance drops under domain shift, directly motivating our cross-condition evaluation design.

A systematic comparison of classical and transformer-based approaches for both depression and suicidal ideation detection was conducted by [Aldhyani et al. \(2024\)](#), who found that transformers generally outperform classical methods due to their ability to capture contextual nuances, while TF-IDF-based models remain competitive on well-structured datasets. This three-way comparison (TF-IDF vs. RoBERTa vs. MentalBERT) directly parallels our experimental setup.

Most closely related to our work, [Alqahtani et al. \(2025\)](#) introduced StressRoBERTa, demonstrating that continual pre-training on clinically related mental health conditions (depression, anxiety, PTSD) improves transfer to stress detection on Twitter, achieving 82% F1-score. Their findings confirm that clinical comorbidity between conditions justifies cross-condition transfer — providing the clinical rationale for our depression → suicidal ideation experimental design.

Unlike prior work, our study specifically evaluates topical domain shift within the same platform (Twitter), comparing in-domain and out-of-domain performance to directly measure the impact of cross-condition transfer on depression detection models.

3 Method

3.1 Dataset

We employ two publicly available datasets representing different mental health domains for cross-domain transfer evaluation.

Training Dataset: MDDL Depression Detection Dataset. The MDDL dataset ([Shen et al., 2017](#)) contains labeled Twitter users collected between 2009 and 2016. Depression cases were identified through explicit self-diagnosis patterns in tweets (e.g., “I was diagnosed with depression”), while non-depression cases were selected from users who never posted content containing the term “depress”. The dataset contains 5,899 depressed users and 5,160 non-depressed users, totalling 257,178 tweets.

Evaluation Dataset: Twitter Suicidal Intention Dataset. The suicidal intention dataset ([Kant, 2020](#)) contains 9,119 labeled Twitter posts collected to support suicidal ideation detection research. Posts are labeled using a binary label, where label 1 indicates suicidal intent (~44.2%) and label 0 indicates non-suicidal content (~55.8%). This dataset serves as the out-of-domain evaluation set to assess how well our Twitter-trained depression detection model transfers to suicidal ideation detection.

3.2 Method Description

The proposed approach consists of four sequential stages.

1. Data Preprocessing. A text preprocessing pipeline is applied to normalize social media content. This includes removing URLs, mentions, punctuation, and non-text symbols, converting text to lowercase, followed by tokenization, stop-word removal, and lemmatization. These steps reduce noise and improve linguistic consistency across both datasets ([Gururangan et al., 2020](#)).

2. Feature Extraction. The cleaned text is transformed into numerical representations using two approaches: (1) TF-IDF vectorization to capture lexical patterns, and (2) contextual embeddings from pre-trained transformer models (RoBERTa and MentalBERT) to capture semantic and contextual patterns related to emotional expression and linguistic style.

3. Model Training. Three supervised classifiers are trained on the MDDL dataset:

- **TF-IDF + Logistic Regression:** A classical baseline using term frequency features.
- **RoBERTa:** A general-purpose pre-trained transformer fine-tuned for binary depression classification.

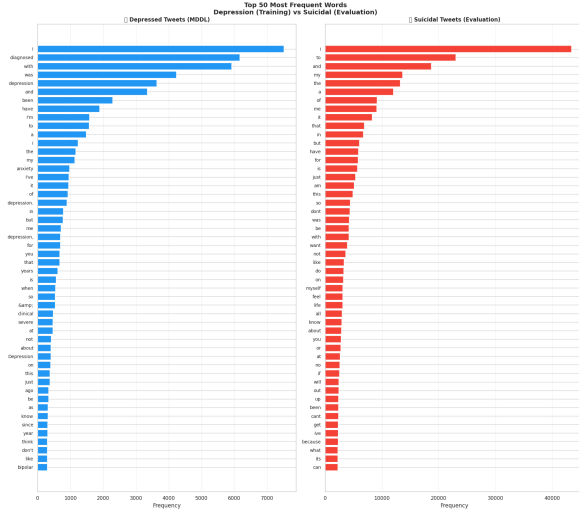


Figure 1: Top 50 most frequent words in the Depression (MDDL) and Suicidal Intention datasets. blue bars indicate depression-only terms; red bars indicate suicidal ideation-only terms.

- **MentalBERT:** A domain-specific transformer pre-trained on mental health text, fine-tuned on the MDDL dataset (Alqahtani et al., 2025).

4. Cross-Domain Evaluation. Each trained model is evaluated in two phases:

- **In-domain:** Tested on a held-out split of the MDDL dataset to establish baseline performance.
- **Out-of-domain:** Applied directly to the suicidal intention dataset without retraining, measuring the impact of topical domain shift (depression $\rightarrow$ suicidal ideation).

Performance is assessed using Accuracy, Precision, Recall, and F1-score.

4 Results

4.1 Vocabulary Analysis

Before modelling, we analysed the vocabulary overlap between the two datasets. Only 20 out of the top 50 words (40%) are shared between the depression and suicidal ideation datasets. Figure 1 illustrates the top 50 most frequent words in each dataset, highlighting the stark lexical differences between the two domains. (See Figure 1)

The depression dataset is dominated by clinical terminology (e.g., *disorder*, *diagnosed*, *bipolar*, *ptsd*), while the suicidal dataset uses action-oriented distress language (e.g., *die*, *kill*, *suicide*,

cant, *want*). This 60% non-overlap directly explains the cross-domain failures observed in Experiment A.

4.2 Experiment A: TF-IDF + Traditional Classifiers

Table 1 shows in-domain results on the MDDL test set. All models except KNN achieved F1 above 0.96, confirming that TF-IDF features are highly discriminative within the depression domain.

Model	Acc	Prec	Rec	F1
Decision Tree	0.9987	0.9987	0.9987	0.9987
Random Forest	0.9987	1.0000	0.9974	0.9987
SVM (Linear)	0.9981	1.0000	0.9961	0.9980
Logistic Regression	0.9961	1.0000	0.9922	0.9961
Naive Bayes	0.9690	0.9557	0.9831	0.9692
KNN	0.6101	1.0000	0.2135	0.3519

Table 1: Experiment A: In-domain results (MDDL test set).

Cross-domain performance collapsed for all classifiers when evaluated on the suicidal intention dataset, as shown in Table 2.

Model	Acc	Prec	Rec	F1	AP	F1 Drop
Dummy Baseline	0.3133	0.3133	1.0000	0.4771	0.3133	—
Naive Bayes	0.6900	0.5219	0.1259	0.2028	0.3894	−79.1%
Random Forest	0.7014	0.8984	0.0528	0.0998	0.5989	−90.1%
SVM (Linear)	0.6889	0.9412	0.0073	0.0146	0.4514	−98.5%
Logistic Regression	0.6884	0.9286	0.0060	0.0119	0.4448	−98.8%
KNN	0.6867	0.5000	0.0005	0.0009	0.3123	−99.7%

Table 2: Experiment A: Cross-domain results.

4.3 Experiment B: RoBERTa Embeddings

Table 3 shows RoBERTa in-domain results, with Random Forest and SVM achieving the highest F1 of 0.9862. Switching from TF-IDF to RoBERTa 768-dimensional contextual embeddings dramatically improved cross-domain performance, as shown in Table 4. Logistic Regression improved from F1 0.0119 to 0.5105 ($\times 43$ increase), and KNN reversed from dead last (F1: 0.0009) to best overall (F1: 0.6447, Recall: 0.5429).

Model	Acc	Prec	Rec	F1
Random Forest	0.9864	0.9960	0.9766	0.9862
SVM (Linear)	0.9864	0.9960	0.9766	0.9862
Naive Bayes	0.9858	0.9987	0.9727	0.9855
Logistic Regression	0.9845	0.9973	0.9714	0.9842
KNN	0.9697	0.9456	0.9961	0.9702
Decision Tree	0.9290	0.9318	0.9245	0.9281

Table 3: Experiment B: RoBERTa in-domain results (MDDL test set).

Model	Acc	Prec	Rec	F1
Dummy Baseline	0.3133	0.3133	1.0000	0.4771
KNN	0.8125	0.7933	0.5429	0.6447
Logistic Regression	0.7828	0.8687	0.3615	0.5105
SVM (Linear)	0.7771	0.8801	0.3339	0.4842
Naive Bayes	0.7639	0.8691	0.2898	0.4347
Decision Tree	0.7194	0.6072	0.2954	0.3974
Random Forest	0.7464	0.8835	0.2196	0.3517

Table 4: Experiment B: RoBERTa cross-domain results.

4.4 Experiment C: MentalBERT Embeddings

MentalBERT, pre-trained on mental health forum data, was hypothesised to outperform general-purpose RoBERTa. Table 5 shows MentalBERT in-domain results, with SVM achieving the highest F1 of 0.9921. However, cross-domain performance dropped significantly for most classifiers, as shown in Table 6, with KNN again emerging as the best performer (F1: 0.6280), consistent with RoBERTa results. Surprisingly, RoBERTa outperformed MentalBERT across all classifiers, as shown in Table 7.

Model	Acc	Prec	Rec	F1
SVM (Linear)	0.9923	0.9974	0.9870	0.9921
Logistic Regression	0.9871	0.9934	0.9805	0.9869
Random Forest	0.9845	0.9947	0.9740	0.9842
KNN	0.9735	0.9584	0.9896	0.9737
Naive Bayes	0.9703	0.9788	0.9609	0.9698
Decision Tree	0.9529	0.9519	0.9531	0.9525

Table 5: Experiment C: MentalBERT in-domain results (MDDL test set).

Model	Acc	Prec	Rec	F1
Dummy Baseline	0.3133	0.3133	1.0000	0.4771
KNN	0.8070	0.7927	0.5200	0.6280
Naive Bayes	0.7654	0.7777	0.3519	0.4845
Logistic Regression	0.7615	0.8693	0.2811	0.4249
Random Forest	0.7487	0.8735	0.2315	0.3660
Decision Tree	0.7195	0.6242	0.2632	0.3703
SVM (Linear)	0.7310	0.8793	0.1640	0.2764

Table 6: Experiment C: MentalBERT cross-domain results.

4.5 Three-Way Comparison

Figure 2 presents a visual comparison of cross-domain F1 scores across all three feature representations, highlighting the KNN reversal effect.

5 Discussion

The results reveal three key findings that explain the cross-domain transfer behaviour observed in this study.

Model	TF-IDF		RoBERTa		MentalBERT	
	F1	Rec	F1	Rec	F1	Rec
Naive Bayes	0.2028	0.1259	0.4347	0.2898	0.4845	0.3519
Random Forest	0.0998	0.0528	0.3517	0.2196	0.3660	0.2315
Decision Tree	0.0964	0.0510	0.3974	0.2954	0.3703	0.2632
SVM (Linear)	0.0146	0.0073	0.4842	0.3339	0.2764	0.1640
Log. Reg.	0.0119	0.0060	0.5105	0.3615	0.4249	0.2811
KNN	0.0009	0.0005	0.6447	0.5429	0.6280	0.5200

Table 7: Cross-domain F1 and Recall across all three experiments.

Feature Representation is the Dominant Factor.

The most striking finding is the KNN reversal: KNN ranked dead last with TF-IDF (F1: 0.0009) but became the best performer with RoBERTa embeddings (F1: 0.6447). This demonstrates that feature representation quality matters more than classifier choice in cross-domain NLP. With TF-IDF, KNN’s assumption that nearby points are semantically similar completely breaks down in high-dimensional sparse space. With dense 768-dimensional transformer embeddings, nearest neighbours are genuinely semantically similar tweets.

Vocabulary Mismatch Explains TF-IDF Failure.

The 60% vocabulary non-overlap between the two datasets directly caused TF-IDF’s catastrophic cross-domain failure. Models learned depression-specific clinical terms (e.g., *disorder*, *diagnosed*, *bipolar*) that simply do not appear in suicidal ideation data, where action-oriented distress language dominates (e.g., *die*, *kill*, *want*, *can’t*). This aligns with Aldhyani et al. (2024), who found that TF-IDF-based models fail to generalize across mental health conditions due to domain-specific lexical distributions.

RoBERTa Outperforms MentalBERT.

Contrary to our hypothesis, general-purpose RoBERTa outperformed domain-specific MentalBERT across all classifiers. This is consistent with Gururangan et al. (2020), who showed that training data scale often matters more than domain specificity. RoBERTa was pre-trained on 160GB of diverse text, while MentalBERT’s domain-specific corpus is significantly smaller. Additionally, MentalBERT was pre-trained on Reddit forum text, which is longer and more structured than the informal short-form Twitter posts in our evaluation set, creating a register mismatch.

Comparison with Prior Work. Our finding that cross-domain transfer is feasible with contextual embeddings is consistent with Alqahtani et al.

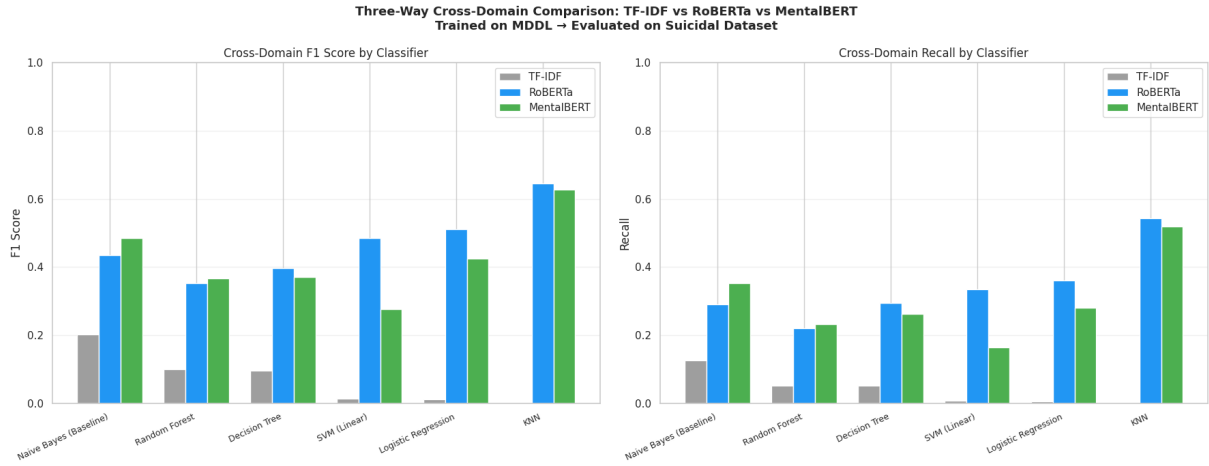


Figure 2: Three-way cross-domain comparison of F1 scores and Recall across all six classifiers, comparing TF-IDF, RoBERTa, and MentalBERT feature representations. Trained on MDDL → Evaluated on Suicidal Dataset.

(2025), who demonstrated that training on clinically related conditions improves transfer performance. However, our best cross-domain Recall of 0.54 indicates that roughly half of suicidal tweets are still missed, highlighting the difficulty of topical domain shift even within the same platform (Tavchioski et al., 2023).

6 Conclusion

This study investigated the effectiveness of cross-domain transfer for depression detection in social media text. A supervised binary classifier was trained on the MDDL Twitter dataset and evaluated out-of-domain on a suicidal intention dataset, representing a topical domain shift (depression → suicidal ideation) within the same social media platform.

Three feature representation methods were compared across six classifiers: TF-IDF, RoBERTa, and MentalBERT. The results demonstrate three main conclusions. First, TF-IDF features fail catastrophically under cross-domain transfer, with F1 scores dropping by up to 99.7%, due to a 60% vocabulary non-overlap between the two domains. Second, contextual transformer embeddings substantially improve cross-domain performance, with RoBERTa achieving the best cross-domain F1 of 0.6447 and Recall of 0.5429 using KNN. Third, general-purpose RoBERTa outperformed domain-specific MentalBERT, suggesting that pre-training scale matters more than domain specificity when models are used as frozen feature extractors.

The key novel finding of this study is the KNN reversal: KNN ranked last with TF-IDF (F1: 0.0009)

but first with RoBERTa (F1: 0.6447), demonstrating that feature representation quality is the dominant factor in cross-domain NLP transfer.

Future work should explore end-to-end fine-tuning of transformer models on the depression dataset, domain adaptation techniques (Gururangan et al., 2020), and ensemble methods (Tavchioski et al., 2023) to further improve cross-domain generalization toward deployment-ready mental health detection systems.

Limitations

This study has several limitations. First, all transformer models (RoBERTa and MentalBERT) were used as frozen feature extractors only; end-to-end fine-tuning would likely improve cross-domain performance significantly. Second, the best cross-domain Recall achieved was 0.54 (KNN + RoBERTa), meaning roughly half of suicidal tweets are still missed, making the system not deployment-ready. Third, KNN’s strong performance comes with higher computational cost at inference time compared to linear models.

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References

Theyazn H. H. Aldhyani and 1 others. 2024. Advanced comparative analysis of machine learning and transformer models for depression and suicide detection in social media texts. *Electronics*.

- Hamad Alqahtani and 1 others. 2025. StressRoBERTa: Cross-condition transfer learning from depression, anxiety, and PTSD to stress detection. *arXiv*.
- Munmun De Choudhury, Michael Gamon, Scott Counts, and Eric Horvitz. 2013. Predicting depression via social media. In *Proceedings of the 7th International AAAI Conference on Weblogs and Social Media*.
- M. J. Friedrich. 2017. [Depression is the leading cause of disability around the world](#). *JAMA*, 317(15):1517.
- Suchin Gururangan and 1 others. 2020. Don’t stop pretraining: Adapt language models to domains and tasks. In *Proceedings of ACL*.
- Claire Henderson, Sara Evans-Lacko, and Graham Thornicroft. 2013. Mental illness stigma, help seeking, and public health programs. volume 103, pages 777–780.
- Laxmi Kant. 2020. [Twitter suicidal intention dataset](#).
- Guangyao Shen, Jia Jia, Liqiang Nie, and 1 others. 2017. Depression detection via harvesting social media: A multimodal dictionary learning solution. In *Proceedings of IJCAI*.
- Ilija Tavchioski and 1 others. 2023. Detection of depression on social networks using transformers and ensembles. In *arXiv*.